

AUDIT RESOLUTION REPORT & CHANGELOG

Verification of compliance fixes, input sanitization wrappers, and API exception handling.

Target System	IDBI MSME Risk Intelligence Engine
Repository	MSME-BANK-PREDICTION
Location	c:/Users/PARTH/OneDrive/Desktop/project/MSME-BANK-PREDICTION
Audit Date	July 7, 2026
Auditors	External Risk Auditor & Portfolio Bank Manager
Compliance Standards	RBI Fair Practices Code, ECOA Fair Lending Guidelines

1. STRUCTURAL RISK ASSESSMENT

An exhaustive review was performed on the XGBoost ML pipeline features and downstream loss projections:

- **Direct Structural Bias:** Checked the 10 leading indicators used in model scoring. All protected classes (Gender, Race, Caste, Age) are completely absent. Features strictly represent credit behavior and leverage ratios.
- **Indirect NLP Bias:** Evaluated officer note embeddings for demographic triggers. The lexicon keyword-matcher fallback was active, guaranteeing identical risk scores regardless of applicant gender or region. However, downstream monitoring is required if the Sentence-Transformer is activated.
- **Sector Capital Allocation:** Construction and Manufacturing sectors carry LGD values of 55% and 45% respectively, elevating Expected Loss (EL) by up to 83% over Technology (30% LGD). This has been flagged as a sector-concentration allocation risk.

2. RESOLVED RUNTIME EXCEPTIONS

The following system-level vulnerabilities were uncovered and fully resolved in the production codebase:

Vulnerability	Root Cause	Remediation
API Exception Crash (KeyError)	predict_pd expected complete dictionary keys. Missing features raised a KeyError in pandas indexing.	Reindexed features to the complete FEATURES schema, filling missing keys with NaN.
Silent Formatting Coercion	Ingestion coerced formatted text (currency symbol, commas, percentages) to NaN, silently replacing them with baseline medians.	Added string cleaning and regex parser wrapper to strip non-numeric tokens prior to conversion.
Negative Parameter Bypass	Negative values bypassed conversion, feeding downstream models and creating an exploit vector (e.g. -25% utilization scored as 1.34% PD).	Applied domain-specific clipping filters (e.g., utilization clipped to [0.0, 1.0]).

Silent Overwrite Bug	Ingesting files missing a single parameter triggered full schema derivation, completely overriding all provided features with medians.	Refactored feature derivation to conditionally write missing values, preserving existing valid inputs.
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3. CODE VALIDATION DIFF COMPARISONS

Modification A: KeyError Resolution in predict_pd

Original Buggy Snippet	Remediated Sanitized Code
<pre>def predict_pd(features: dict) -> float: return float(predict_portfolio(pd.DataFrame([features]))[0])</pre>	<pre>def predict_pd(features: dict) -> float: df = pd.DataFrame([features]) for f in FEATURES: if f not in df.columns: df[f] = np.nan return float(predict_portfolio(df)[0])</pre>

Modification B: Silent Overwrite Fix in _derive_features

Original Buggy Snippet	Remediated Sanitized Code
<pre>out = df.copy() out['revolving_utilization'] = np.clip(util, 0.01, 0.99) out['debt_ratio'] = debt out['late_30_59'] = np.clip(emi, 0, 12).astype(int)</pre>	<pre>out = df.copy() if 'revolving_utilization' not in out.columns or out['revolving_utilization'].isna().all(): out['revolving_utilization'] = np.clip(util, 0.01, 0.99) if 'debt_ratio' not in out.columns or out['debt_ratio'].isna().all(): out['debt_ratio'] = debt</pre>

Modification C: Input Sanitization Helper clean_numeric_series

Remediated Input Sanitizer Helper Wrapper
<pre>def clean_numeric_series(s: pd.Series) -> pd.Series: if s is None or len(s) == 0: return pd.Series(dtype=float) s_str = s.astype(str).str.strip() s_cleaned = s_str.str.replace(r'[\u20B9\u20A8\s,]', '', regex=True) pct_mask = s_cleaned.str.endswith('%', na=False) s_cleaned = s_cleaned.str.rstrip('%') numeric_s = pd.to_numeric(s_cleaned, errors='coerce') if pct_mask.any(): numeric_s = numeric_s.mask(pct_mask, numeric_s / 100.0) return numeric_s</pre>

4. VERIFICATION AUDIT RESULTS

Test Name	Injected Input	Expected Behavior	Auditor Verification Status